



Intelligent Knowledge Lakes: The Age of Artificial Intelligence and Big Data

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Abstract. The continuous improvement in connectivity, storage and data processing capabilities allow access to a data deluge from the big data generated on open, private, social and IoT (Internet of Things) data islands. Data Lakes introduced as a storage repository to organize this raw data in its native format until it is needed. The rationale behind a Data Lake is to store raw data and let the data analyst decide how to curate them later. Previously, we introduced the novel notion of Knowledge Lake, i.e., a contextualized Data Lake, and proposed algorithms to turn the raw data (stored in Data Lakes) into contextualized data and knowledge using extraction, enrichment, annotation, linking and summarization techniques. In this tutorial, we introduce Intelligent Knowledge Lakes to facilitate linking Artificial Intelligence (AI) and Data Analytics. This will enable AI applications to learn from contextualized data and use them to automate business processes and develop cognitive assistance for facilitating the knowledge intensive processes or generating new rules for future business analytics.

Keywords: Knowledge Lake · Data Lake · Data analytics · Data curation · Artificial Intelligence · Big data

1 Introduction

Today, the advancement in Artificial Intelligence (AI) and Data Science has the potential to transform business processes in fundamental ways, by assisting knowledge workers in communicating analysis findings, supporting evidences and to make decisions. Almost every organization is now focused on understanding their business and transform data into actionable insights. For example, governments derive insights from vastly growing private, open and social data for improving government services, such as to personalize the advertisements in elections, improve government services, predict intelligence activities and to improve

national security and public health [6, 12, 16, 28, 32]. In this context, organizing vast amount of data gathered from various private/open data islands will facilitate dealing with a collection of independently-managed datasets (from relational to NoSQL), diversity of formats and non-standard data models.

The notion of a Data Lake [1, 4, 20, 24, 33] has been coined to address this challenge and to convey the concept of a centralized repository containing limitless amounts of raw (or minimally curated) data stored in various data islands. The rationale behind a Data Lake is to store raw data [18] and let the data analyst decide how to cook/curate them later. While Data Lakes, do a great job in organizing big data and providing answers on known questions, the main challenges are to understand the potentially interconnected data stored in various data islands and to prepare them for analytics. In our previous work [5, 6], we introduced the novel notion of Knowledge Lake, i.e., a contextualized Data Lake, and proposed algorithms to turn the raw data (stored in Data Lakes) into contextualized data and knowledge using extraction, enrichment, annotation, linking and summarization techniques [11, 13, 15]. In particular, a Knowledge Lake is a centralized repository containing virtually inexhaustible amounts of both data and contextualized data that is readily made available anytime to anyone authorized to perform analytical activities. The Knowledge Lake will provide the foundation for big data analytics by automatically curating [8, 9] the raw data in the Data Lake and to prepare them for deriving insights.

In this tutorial, we introduce Intelligent Knowledge Lakes to facilitate linking Artificial Intelligence (AI) and Data Analytics. This will enable AI applications to learn from contextualized data and use them to automate business processes and develop cognitive assistance for facilitating the knowledge intensive processes or generating new rules for future business analytics [3, 12]. We focus on motivating scenarios in policing [27], budget [6] and health [7]; and present how Intelligent Knowledge Lakes can assist knowledge workers in knowledge intensive processes to deal with big data and dig for facts in an easy way.

2 From Data to Big Data

Data. In computing, data is information that has been translated into a form that is efficient for *storage* and *processing*. Data can be generated on various islands of data, from social (e.g., on Twitter [26] and Facebook [23]) and Internet of Things [34] (IoT, e.g., CCTVs and smart cars) to private (e.g., personal and business data) and open (e.g., news and Websites) data islands. For example, when we take a photo using our cell phone, open a banking account, or post a news on a social media (e.g., on Twitter [26] or LinkedIn [31]), we are generating data. Even - when we walk, exercise in gym or watching TV - smart devices such as smart phones, watches and TVs can trace our location, health signs and interests to generate large amount of data. From the storage point of view, data can be structured (e.g., a customer record in a banking system), semi-structured (e.g., a Tweet in Twitter) or unstructured (e.g., a document, audio or video file). Common data processing operations include organizing, curating, analyzing and visualizing.

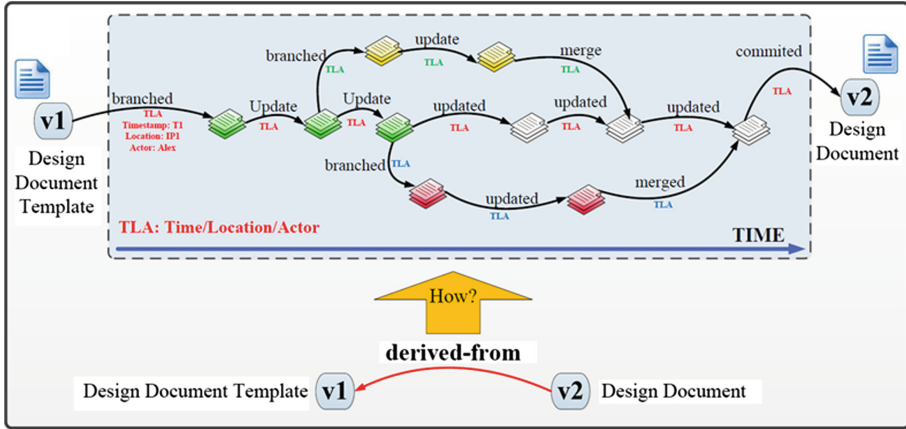


Fig. 1. Provenance, a kind of metadata, refers to the documentation of an object’s lifecycle. This documentation (often represented as a graph) should include all the information necessary to reproduce a certain piece of data or the process that led to it [14, 25].

Metadata. Metadata is the data that provides information about other data. The goal is to help users properly understand the data. Metadata (e.g., descriptive, structural and administrative metadata) can be generated manually and/or automatically. Tracing is a mechanism to generate metadata. For example, when we use our cell phone, a set of algorithms can trace our movement (location services), what Apps we are using and how frequently. Or, a social network App can trace the activities of a user, including: which days of week and what time of the days the user is more active, what sort of content the user liked or reshared, and what type of people (e.g., from personality, behavior and attitude view points) the user is following. Metadata can also help in generating the story of a digital object from now back to its derivation, i.e., provenance [14, 25] (the documentation of an object’s lifecycle), including: who created the file, on which system (e.g., IP address) and when, how the digital object evolved over time and more. Figure 1 illustrates an example where provenance can help understanding how a document evolve over time [14].

Big Data. Big data [22, 35] is different from a large dataset. For example, we have had super computers to deal with processing and analyzing a large data set. However, big data can be seen as a massive number of small data islands from Private (Personal/Business), Open, Social and IoT Data [3, 11]. Typical properties of the big data include: wide physical distribution, diversity of formats, non-standard data models, independently-managed and heterogeneous semantics. Organizing, curating, analyzing and presenting this data is challenging and of high interest. For examples, consider a motivating scenario in policing, in investigation processes such as Boston Marathon bombing¹, the police investi-

¹ https://en.wikipedia.org/wiki/Boston_Marathon_bombing.



Fig. 2. An example of the big data problem in police investigation [27]. Big data can be seen as a massive number of small data islands from Private (Personal/Business), Open, Social and data islands [3, 11].

gator needs to understand the big data generated on various private, social, open and IoT data islands; analyze them and link them to police historical data. This is quite important, as modern police investigation processes are often extremely complex, data-driven and knowledge-intensive. Linking big data to process execution, will facilitate communicating analysis findings, supporting evidences and to make decisions. Figure 2 illustrates a big data problem scenario in policing [27].

3 Data Curation: From Ingestion and Cleaning to Extraction and Enrichment

One of the main challenges in understanding big data is to transform raw data into actionable insights. To achieve this goal it will be vital to prepare and curate the raw data for analytics. Data curation has been defined as the active and ongoing management of data through its lifecycle of interest and usefulness [17, 29]. Data curation includes all the processes needed for principled and controlled data creation, maintenance, and management, together with the capacity to add value to data. Big data curation should involve [6, 21, 22, 29]:

- Identifying Relevant Data Islands: with more data repositories constantly being published every day, choosing appropriate data sources for a specific analyst goal becomes very important.
- Ingesting Data and Knowledge: data ingestion is the process of obtaining and importing data for immediate use or storage in a database. Data can be streamed in real time or ingested in batches.

- Cleaning: also known as data cleansing and data scrubbing, is the process of amending or removing data in a database that is incorrect, incomplete, improperly formatted or duplicated.
- Integration: data integration combines data from multiple sources. Issues during data integration include: schema integration, detecting and resolving data value conflicts, and removing duplicates and redundant data.
- Transformation (normalization and aggregation): data transformation is usually used to smooth the noisy data, summarize, generalize, or normalize the data scale falls within a small, specified range. The process may include: smoothing (remove noise from data, e.g., binning, clustering, regression), normalization (scaled to fall within a small, specified range such as -1.0 to 1.0 or 0.0 to 1.0), feature construction (e.g., new attributes constructed or added from the given ones), aggregation (e.g., summarization or aggregation operations apply to data) and generalization (e.g., concept hierarchy climbing. For example, low level/ primitive/raw data are replace by higher level concepts).
- Adding Value: a very important step in the curation process which includes: extraction, enrichment, annotation, linking (to external knowledge sources and services) and summarization tasks. Figure 3 illustrates an example of adding value to a raw tweet [9].

4 From Data Lakes to Knowledge Lakes

Organizing vast amount of raw data gathered from various data islands, i.e., *Data Lake* [1,4,33], will facilitate dealing with a collection of independently-managed datasets in private (personal and business data), open (news and Websites), social (such as Twitter, Facebook and LinkedIn) and IoT (smart entities such as CCTVs and smart cars) data islands. The notion of a Data Lake has been coined to address this challenge and to convey the concept of a centralized repository containing limitless amounts of raw (or minimally curated) data stored in various data islands. The rationale behind the Data Lake is to store raw data and let the data analyst decide how to cook/curate them later.

In our previous work [5,6], we introduced the notion of *Knowledge Lake*, i.e. a contextualized Data Lake, as a centralized repository containing virtually inexhaustible amounts of both data and *contextualized data* that is readily made available anytime to anyone authorized to perform analytical activities. The term *Knowledge* here refers to a set of facts, information, and insights extracted from the raw data using data curation techniques used to transfer an Information-Item into a Featurized-, Semantic- and Contextualized-Items [6]. The Knowledge Lake will provide the foundation for big data analytics by automatically curating the raw data in the Data Lake and to prepare them for deriving insights. For the raw information items stored in the Lake, we provide services to automatically [15]: extract features, enrich them, link them to external knowledge sources, and annotate them. On top of the Knowledge Lake, we provide a single federated search service which enables the analyst querying the raw data (in the Data

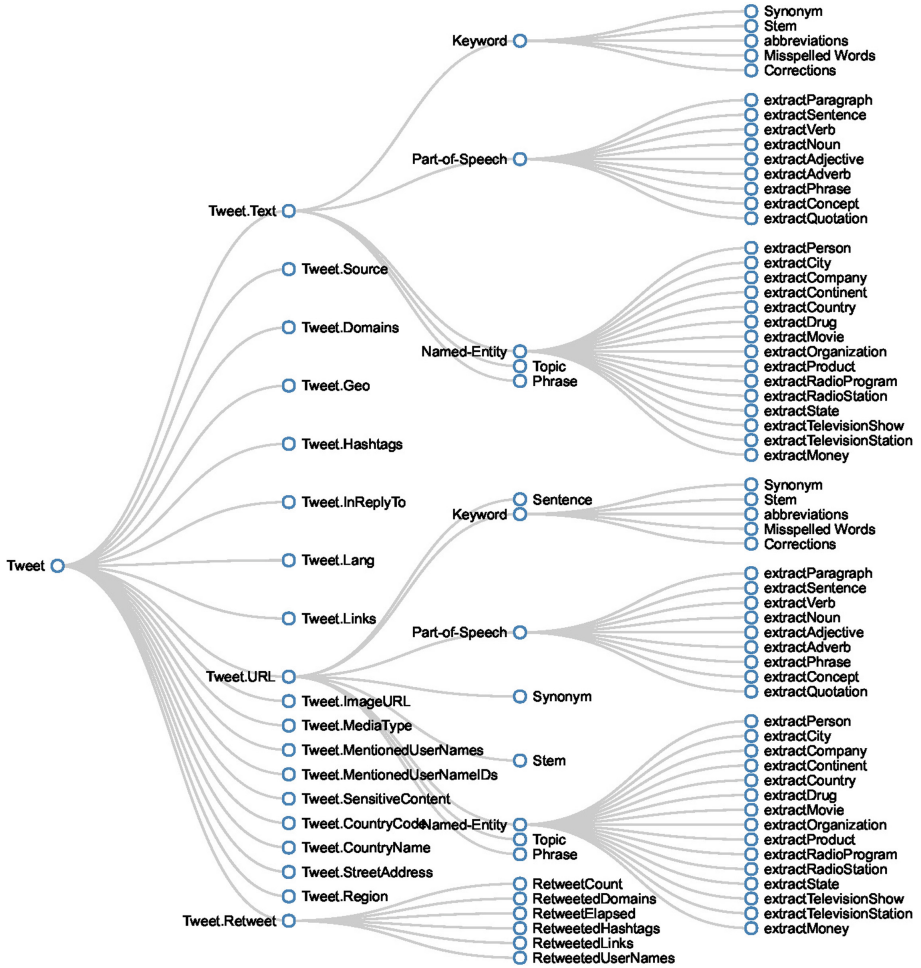


Fig. 3. An example of an automatically curated tweet [9]. Data Curation is the process of transforming raw data into contextualized data and knowledge. The first step is to extract features from raw data, e.g., extract keywords or name entities from the text of the tweet or extract objects from an image mentioned in the tweet [13, 15]. The next step is to enrich the extracted features, annotate them and link them to external knowledge sources and services [6]. For example, to enrich an extracted keyword with synonyms (e.g., using wordnet [30]) and similar keywords (e.g., using word2vec [19]) or link an extracted named entity to Wikidata (wikidata.org/).

Lake) as well as the contextualized items in the Knowledge Lake using Full-text Search, SQL and SPARQL. Figures 4 and 5 illustrate the architecture and the main components of the Data Lake and Knowledge Lake respectively.

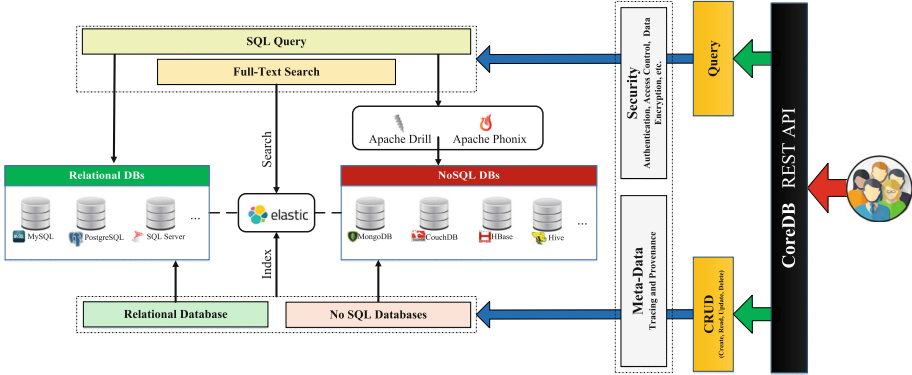


Fig. 4. Data Lake Architecture [4].

5 AI-enabled Knowledge Lakes

Today, the advancement in Artificial Intelligence (AI) and Data Science is rapidly transforming how businesses operate. In this context, Knowledge Lakes can help organization understand their big data, use it in a smart way, and feed it into their AI engine. Figure 6 illustrates an AI-enabled pipeline to facilitate the decision-making process.

The first step in the pipe line is to assist analysts to deal with a collection of datasets, from relational to NoSQL, that holds a vast amount of data gathered from various data islands, i.e., Data Lake. To address this challenge, we leverage our previous work [4], CoreDB: a Data Lake as a Service, to identify (IoT, Private, Social and Open) data sources and ingest the big data in the Data Lake. CoreDB manages multiple database technologies (from relational to NoSQL), offers a built-in design for security and tracing, and provides a single REST API to organize, index and query the data and metadata in the Data Lake.

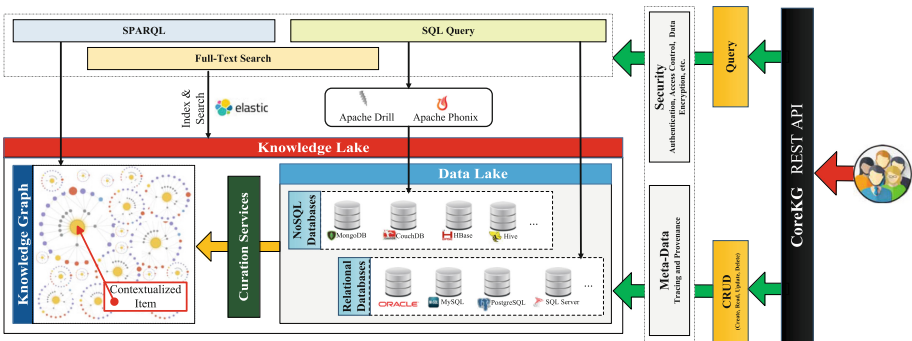


Fig. 5. Knowledge Lake Architecture [5].

Next, we provide services [15] to automatically curating the raw data in the Data Lake and to prepare them for deriving insights. To address this challenge, we leverage our previous work [5], CoreKG: a contextualized Data Lake, to provide the foundation for big data analytics. The data curation APIs [15] in the Knowledge Lake provide curation tasks such as extraction, linking, summarization, annotation, enrichment, classification and more. This will enable us to add features - such as extracting keyword, part of speech, and named entities such as Persons, Locations and Organizations; providing synonyms and stems for extracted information items leveraging lexical knowledge bases for the English language such as WordNet [30]; linking extracted entities to external knowledge bases such as Google Knowledge Graph² and Wikidata³; discovering similarity among the extracted information items; classifying, indexing, sorting and categorizing data - into the data and knowledge persisted in the Knowledge Lake.

The data and knowledge in the knowledge lake will form a large graph, i.e. Knowledge Graph, of entities⁴ and relationships among them. Entities can be simple such as a raw data (e.g. a document). Entities can be atomic such as a keyword and topic extracted from a document. The major challenges here are: (i) how to extend decision support on multidimensional networks, considering both data objects and the relationships among them; and (ii) providing multiple views at different granularities is subjective: depends on the perspective of analysts how to partition graphs and apply further operations on top of them.

To achieve this goal, we present novel techniques (including our work on graph OLAP [11]) for intelligent narrative discovery to summarize the large amount of curated data in the Knowledge Lake and to transform a Knowledge Lake into a smart and intelligent lake. Figure 6 illustrates a pipeline to leverage Domain Experts and Crowd Knowledge (e.g., using feedback, surveys, interviews etc. to build a domain specific knowledge base) and use that knowledge to summarize the large knowledge graph (in a Knowledge Lake) and enable intelligent narrative discovery. This will enable AI applications to learn from contextualized data, link them to domain knowledge and use them to automate business processes and develop cognitive assistance for facilitating the knowledge intensive processes or generating new rules for future business analytics [3, 12].

In this tutorial, we present three motivating scenarios from our recent work:

- DataSynapse [6, 10]: we present a scenario for analyzing Urban Social Issues from Twitter as it relates to the Government Budget: it is paramount to stabilize the economy through timely and dynamic adjustment in expenditure plans by considering related social issues such as a problem or conflict raised by society ranging from local to national issues such as Health, Social Security, Public Safety, Welfare Support and Domestic Violence.
- personality2vec [7]: we discuss how we design intelligent Knowledge Lakes extended with deep learning techniques to analyze and summarize social data

² <https://developers.google.com/knowledge-graph>.

³ <https://www.wikidata.org/>.

⁴ An entity E is represented as a data object that exists separately and has a unique identity. Entities are described by a set of attributes.

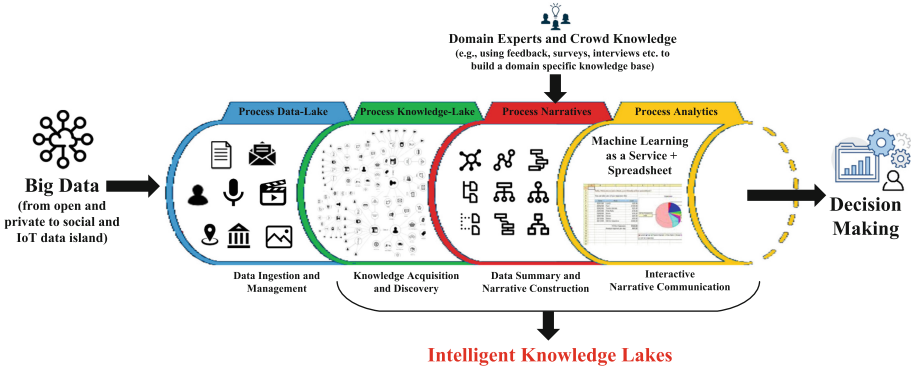


Fig. 6. Intelligent Knowledge Lakes.

to predict patterns of behavioral disorders (e.g., criminal and extremist behaviors) on social networks.

- iCOP [8, 27]: we present a cognitive assistant, namely iCOP, that we developed to help police investigators identify evidences from the Knowledge Lake in an easy way. In iCOP, we proposed novel techniques to summarize the Knowledge Lake to construct process narratives and enable the analysis of narratives and dig for facts in an easy way, using smart sheets [2].

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References

1. Alsubaiee, S., et al.: Storage management in AsterixDB. *Proc. VLDB Endow.* **7**(10), 841–852 (2014)
2. Amouzgar, F., Beheshti, A., Ghodratinama, S., Benatallah, B., Yang, J., Sheng, Q.Z.: iSheets: a spreadsheet-based machine learning development platform for data-driven process analytics. In: Liu, X., et al. (eds.) *ICSOC 2018*. LNCS, vol. 11434, pp. 453–457. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-17642-6_43
3. Beheshti, A., Benatallah, B., Motahari-Nezhad, H.R.: ProcessAtlas: a scalable and extensible platform for business process analytics. *Softw.: Pract. Exp.* **48**(4), 842–866 (2018)
4. Beheshti, A., Benatallah, B., Nouri, R., Chhieng, V.M., Xiong, H., Zhao, X.: CoreDB: a data lake service. In: *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management, CIKM 2017, Singapore, 06–10 November 2017*, pp. 2451–2454 (2017)
5. Beheshti, A., Benatallah, B., Nouri, R., Tabebordbar, A.: CoreKG: a knowledge lake service. *PVLDB* **11**(12), 1942–1945 (2018)
6. Beheshti, A., Benatallah, B., Tabebordbar, A., Motahari-Nezhad, H.R., Barukh, M.C., Nouri, R.: DataSynapse: a social data curation foundry. *Distrib. Parallel Databases* **37**(3), 351–384 (2019)

7. Beheshti, A., Moraveji-Hashemi, V., Yakhchi, S., Motahari-Nezhad, H.R., Ghafari, S.M., Yang, J.: personality2vec: enabling the analysis of behavioral disorders in social networks. In: Proceedings of the 13th ACM International Conference on Web Search and Data Mining, WSDM 2020, Houston, Texas, USA (2020)
8. Beheshti, A., et al.: iProcess: enabling IoT platforms in data-driven knowledge-intensive processes. In: Weske, M., Montali, M., Weber, I., vom Brocke, J. (eds.) BPM 2018. LNBIP, vol. 329, pp. 108–126. Springer, Cham (2018). https://doi.org/10.1007/978-3-319-98651-7_7
9. Beheshti, A., Vaghani, K., Benatallah, B., Tabebordbar, A.: CrowdCorrect: a curation pipeline for social data cleansing and curation. In: Mendling, J., Mouratidis, H. (eds.) CAiSE 2018. LNBIP, vol. 317, pp. 24–38. Springer, Cham (2018). https://doi.org/10.1007/978-3-319-92901-9_3
10. Beheshti, S., Benatallah, B., Motahari-Nezhad, H.R.: Galaxy: a platform for explorative analysis of open data sources. In: Proceedings of the 19th International Conference on Extending Database Technology, EDBT 2016, Bordeaux, France, 15–16 March 2016, pp. 640–643 (2016)
11. Beheshti, S., Benatallah, B., Motahari-Nezhad, H.R.: Scalable graph-based OLAP analytics over process execution data. *Distrib. Parallel Databases* **34**(3), 379–423 (2016)
12. Beheshti, S., et al.: Process Analytics - Concepts and Techniques for Querying and Analyzing Process Data. Springer, Cham (2016). <https://doi.org/10.1007/978-3-319-25037-3>
13. Beheshti, S., Benatallah, B., Venugopal, S., Ryu, S.H., Motahari-Nezhad, H.R., Wang, W.: A systematic review and comparative analysis of cross-document coreference resolution methods and tools. *Computing* **99**(4), 313–349 (2017)
14. Beheshti, S., Motahari Nezhad, H.R., Benatallah, B.: Temporal provenance model (TPM): model and query language. *CoRR*, abs/1211.5009 (2012)
15. Beheshti, S., Tabebordbar, A., Benatallah, B., Nouri, R.: On automating basic data curation tasks. In: Proceedings of the 26th International Conference on World Wide Web Companion, Perth, Australia, 3–7 April 2017, pp. 165–169 (2017)
16. Berners-Lee, T.: Designing the web for an open society. In: Proceedings of the 20th International Conference on World Wide Web, WWW 2011, Hyderabad, India, 28 March–1 April 2011, pp. 3–4 (2011)
17. Freitas, A., Curry, E.: Big Data Curation. In: Cavanillas, J.M., Curry, E., Wahlster, W. (eds.) *New Horizons for a Data-Driven Economy*, pp. 87–118. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-21569-3_6
18. Gitelman, L.: *Raw Data Is an Oxymoron*. MIT Press, Cambridge (2013)
19. Goldberg, Y., Levy, O.: word2vec explained: deriving Mikolov et al.’s negative-sampling word-embedding method. *arXiv preprint arXiv:1402.3722* (2014)
20. Hai, R., Geisler, S., Quix, C.: Constance: an intelligent data lake system. In: Proceedings of the 2016 International Conference on Management of Data, pp. 2097–2100. ACM (2016)
21. Lord, P., Macdonald, A., Lyon, L., Giarretta, D.: From data deluge to data curation. In: Proceedings of the UK e-science All Hands meeting, pp. 371–375. Citeseer (2004)
22. McAfee, A., Brynjolfsson, E., Davenport, T.H., Patil, D., Barton, D.: Big data: the management revolution. *Harv. Bus. Rev.* **90**(10), 60–68 (2012)
23. Miller, D.: *Tales from Facebook*. Polity, Cambridge (2011)
24. Miloslavskaya, N., Tolstoy, A.: Big data, fast data and data lake concepts. *Procedia Comput. Sci.* **88**, 300–305 (2016)

25. Moreau, L., et al.: The open provenance model core specification (v1.1). *Future Gener. Comput. Syst.* **27**(6), 743–756 (2011)
26. Murthy, D.: *Twitter*. Polity Press, Cambridge (2018)
27. Schiliro, F., et al.: iCOP: IoT-enabled policing processes. In: Liu, X., et al. (eds.) *ICSOC 2018*. LNCS, vol. 11434, pp. 447–452. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-17642-6_42
28. Shadbolt, N., et al.: Linked open government data: Lessons from data.gov.uk. *IEEE Intell. Syst.* **27**(3), 16–24 (2012)
29. Stonebraker, M., et al.: Data curation at scale: the data tamer system. In: *CIDR* (2013)
30. Strapparava, C., Valitutti, A., et al.: Wordnet affect: an affective extension of wordnet. In: *Lrec*, vol. 4, pp. 40. Citeseer (2004)
31. Sumbaly, R., Kreps, J., Shah, S.: The big data ecosystem at linkedin. In: *Proceedings of the 2013 ACM SIGMOD International Conference on Management of Data*, pp. 1125–1134. ACM (2013)
32. Tene, O., Polonetsky, J.: Big data for all: privacy and user control in the age of analytics. *Nw. J. Tech. Intell. Prop.* **11**, xxvii (2012)
33. Terrizzano, I.G., Schwarz, P.M., Roth, M., Colino, J.E.: Data wrangling: the challenging journey from the wild to the lake. In: *CIDR* (2015)
34. Xia, F., Yang, L.T., Wang, L., Vinel, A.: Internet of things. *Int. J. Commun. Syst.* **25**(9), 1101 (2012)
35. Zomaya, A.Y., Sakr, S.: *Handbook of Big Data Technologies*. Springer, Cham (2017). <https://doi.org/10.1007/978-3-319-49340-4>